VR/MR. "Targeting applications addressing unique value propositions of µLED displays is very important," explains Xiaoxi. Hence, the extent to which µLED displays can be beneficial depends on the required application, size, and other factors. Let's have a look at a few.

In residential TV panels. South Korean and Chinese manufacturers are battling with announcements for new ultra-large residential displays. For instance, Samsung's The Wall can max out from 2.8 metres (110 inches) to as big as 14.8 metres (583 inches) horizontally. The µLEDs bring superior colour purity and wider colour gamut than conventional LED displays. Its vibrant colours and deep black levels deliver intense contrast and immaculate detail for a one-of-a-kind visual experience.

The company says it can be sized as large as you want in 4K, or up to 7.4 metres (292 inches) in 8K, for an incredibly immersive viewing experience. In 8K, it can nearly be 15 times larger than a conventional 1.9-metre (75-inch) TV. Also, it claims to offer robust durability and energy efficiency with µLEDs that last for 100,000 hours.

µLED panels look similar to any of the existing regular LED and liquid crystal display (LCD), but the outcome differs significantly in terms of quality. The recently launched televisions come with thousands of µLEDs embedded in their back panel, but in a different arrangement with backlit techniques for efficient dimming control that produces a good contrast ratio. These do not have burn-out issues as in the case of earlier variants.

Fig. 2: MicroLEDs in smart glasses (Source: https://plesseysemiconductors.com)

In your vehicle. Automotive market is benefiting from new applications of the emerging technology. The trend of developing automotive displays has reached another level where larger screens are made with higher resolution and high level of interactivity. Car manufacturers are now replacing the centre console with a touchscreen µLED display.

Typical vehicle displays for automobiles and spacecraft need central cluster panels and head-up display systems. This requires a technology with high reliability, sunlight readability and, most importantly, a wide working temperature. The µLEDs and mini LEDs show great advantage in luminance, lifetime, and robustness even in extreme weather. All the more, the head-up display systems need peak brightness that can be achieved with µLEDs.

In your accessories. Smart watches and eyewear with augmented reality/virtual reality (AR/VR) have been the goal of big players like Google and Apple. Many other companies across the globe are working on small µLED displays that can be fixed in these smart wearables, including headsets and wristbands.

Precisely, these displays require fast moving picture response time (MPRT) and low power consump-

TABLE 1

COMPARISON OF µLEDs WITH OTHER TECHNOLOGIES

	LED-backlit LCD	Mini-LED	OLED	MicroLED
Display type	Backlit	Backlit	Emissive	Emissive
Contrast	Low to medium	Medium	High	High
Response time	Low (ms)	Low (ms)	High (µm)	Very high (ns)
Power efficiency	Medium	Medium-high	Medium	High
Only lit pixels draw power	No	No	Yes	Yes
Lifetime	Long	Long	Medium	Long
Sunlight visibility	Medium	Medium	Medium	High
Operating temperature	~ -20 to 80°C	~ -20 to 80°C	~ -30 to 70°C	~ -100 to 120°C
Need for encapsulation	No	No	Yes	No
Brightness	Medium	Medium	Low-medium	High
Flexibility	Low	Low	High	Medium-high
Viewing angles	Low	Low	High	High
Tech maturity	High	Medium-high	Medium-high	Low

Note: The values in the table are approximate
(Source: The MicroLED Handbook)